Sharecoin: A Bitcoin-Derived Blockchain Providing Verifiable Randomness via GPU-Based Proof-of-Work

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Abstract

Sharecoin is a Bitcoin-Core-derived blockchain that repurposes GPU-based Proof-of-Work (PoW) as a decentralized randomness beacon. By replacing Bitcoin's SHA-256d PoW with the memory-hard KawPow algorithm, Sharecoin exposes the per-block `mix_hash` as an unpredictable, economically expensive, and publicly verifiable source of entropy. The protocol aggregates multiple consecutive blocks to mitigate last-revealer bias and provides a consensus-secured randomness primitive accessible through RPC. Sharecoin preserves full Bitcoin Core compatibility (including P2P networking, validation logic, wallet infrastructure, and RPC semantics) while introducing a novel public-good utility: trustless randomness generation. This whitepaper formally describes the protocol, threat model, implementation details, and applications.

1 Introduction

Randomness is a foundational primitive in distributed systems, cryptography, gaming, lotteries, governance, and decentralized applications. Existing blockchain ecosystems lack a native, trustless randomness source. Smart-contract platforms typically rely on centralized oracles, commit-reveal schemes vulnerable to last-revealer bias, VRF systems requiring trusted key-holders, or external randomness services.

Sharecoin introduces a new approach: mining produces randomness as a public good. By leveraging the inherent unpredictability of GPU-based PoW, Sharecoin transforms the mining process into a decentralized randomness beacon without introducing trusted parties or additional cryptographic assumptions.

Sharecoin is a full Bitcoin Core fork with minimal, auditable modifications. The protocol retains Bitcoin's security model, consensus rules, and network architecture while replacing SHA-256d with KawPow and exposing the resulting entropy.

2 Background and Motivation

2.1 Bitcoin's PoW and Determinism

Bitcoin's SHA-256d PoW is deterministic once a block is mined. The block hash is unpredictable prior to discovery but is not suitable as a randomness beacon because ASIC mining is highly optimized and predictable, miners can cheaply grind nonces, and block hash manipulation is economically feasible.

2.2 GPU-Based PoW as an Entropy Source

Memory-hard PoW algorithms such as KawPow incorporate large memory reads, data-dependent access patterns, GPU architecture variability, and unpredictable mix-state evolution. The `mix_hash` produced during KawPow computation is expensive to influence, unpredictable prior to computation, and infeasible to bias without re-mining the entire block. This makes it a suitable entropy source.

2.3 Design Goals

Sharecoin aims to:

- Provide trustless, verifiable randomness secured by PoW.
- Preserve Bitcoin Core compatibility and security.
- Maintain ASIC resistance and decentralization.
- Offer a transparent, reproducible codebase with minimal diffs.

3 System Architecture

3.1 Consensus Layer

Sharecoin inherits Bitcoin's consensus rules with the following modifications:

- PoW algorithm: KawPow replaces SHA-256d.
- Block header includes `mix_hash`.
- Difficulty adjustment uses LWMA (Linearly Weighted Moving Average), recalculating the target on every block from a weighted average of recent solvetimes, rather than Bitcoin's original fixed 2016-block retargeting window. This reacts within roughly a window's worth of blocks to hashrate changes, relevant for a network with a smaller and more variable miner base than Bitcoin's.

3.2 Mining Process

Mining proceeds as follows:

1. Miner constructs a candidate block.
2. KawPow is executed using the block header and DAG.
3. The algorithm produces:
   - `mix_hash`: internal state hash
   - `result`: final PoW output
4. If `result < target`, the block is valid.
5. `mix_hash` is committed into the block header.

3.3 Randomness Beacon

Sharecoin exposes randomness via:

- Per-block entropy: `mix_hash`
- Aggregated entropy: RPC `getrandombeacon`

The beacon aggregates the last N confirmed blocks:

$$R = H(mix\_hash_i \parallel mix\_hash_{i-1} \parallel \cdots \parallel mix\_hash_{i-N+1})$$

This reduces miner influence and last-revealer bias.

4 Security Analysis

4.1 Threat Model

Adversaries may attempt:

- Biasing randomness by selective block publication.
- Grinding attacks by re-mining blocks.
- 51% attacks to control beacon output.

4.2 Bias Resistance

Biasing requires withholding blocks until a desired `mix_hash` appears. This is economically costly because KawPow is GPU-bound and expensive, withheld blocks lose block rewards, and competitors may publish competing blocks.

4.3 Grinding Resistance

Grinding requires recomputing KawPow repeatedly. Unlike SHA-256d, KawPow is memory-hard, recomputation is slow, and GPU cycles are expensive. Thus grinding is infeasible beyond trivial influence.

4.4 51% Attack Considerations

As with Bitcoin, a 51% attacker can bias randomness, but only at high economic cost and only temporarily. This is identical to Bitcoin's security assumptions.

5 Implementation Details

5.1 Codebase Structure

Sharecoin maintains a minimal diff from Bitcoin Core:

- PoW replaced with KawPow implementation from Ravencoin.
- Block header extended to include `mix_hash`.

- RPC additions: `getrandombeacon`.
- Build system preserved (CMake, vcpkg, autotools).

Patch scripts allow deterministic reproduction of diffs.

5.2 Wallet and Node Software

Sharecoin provides:

- `sharecoind` (daemon)
- `sharecoin-cli` (RPC client)
- `sharecoin-qt` (GUI wallet)
- Windows binaries
- Linux build instructions

5.3 Mining Software

The stratum proxy speaks plain Stratum V1, the same dialect `kawpowminer` uses, and is the only miner confirmed working against the network so far. Other KawPow-capable miners that auto-negotiate a different stratum dialect first (e.g. EthereumStratum/2.0) may fail to connect until they are configured to use plain Stratum V1 directly, if such an option exists in that miner.

- `kawpowminer`

6 Applications

6.1 Decentralized Gaming

Fair randomness for loot drops, matchmaking, and procedural generation.

6.2 Lotteries and Raffles

Trustless draws without centralized operators.

6.3 NFT Reveals

Random trait assignment without oracle dependency.

6.4 DAO Governance

Sortition committees selected via beacon output.

6.5 Scientific Simulations

Entropy source for Monte-Carlo methods requiring public verifiability.

7 Roadmap

7.1 Short-Term

- Formal verification of beacon aggregation.
- Additional RPC endpoints.
- Documentation expansion.

7.2 Medium-Term

- Light-client beacon verification.
- Optional VRF overlay for hybrid randomness.

8 Status and Limitations

Sharecoin is developed and maintained by a single contributor, with no formal team or external funding. The consensus changes described in this paper have not undergone external security audit; all review has been internal. The genesis block reuses Bitcoin's original, provably unspendable coinbase output, so there is no premine and no admin-controlled token allocation. There are currently no exchange listings and no established market for SHC.

9 Conclusion

Sharecoin demonstrates that GPU-based PoW can serve as a practical, trustless randomness beacon. By preserving Bitcoin Core compatibility and introducing a novel entropy primitive, Sharecoin provides a foundation for decentralized applications requiring unbiased randomness without trusted parties or external oracles.